

Chapter 16 Summary: Capacitors

Physics Department | Grade 12 LS/GS

I. Core Definitions

- **Capacitor:** A passive electronic component consisting of two conducting plates (armatures) separated by an insulator (dielectric). It stores electrical energy in an electrostatic field.
- **Charge (Q):** Defined as the quantity of electricity on the positive plate. Due to the principle of conservation of charge: $q_{\text{positive}} = |q_{\text{negative}}| = Q$.
- **Capacitance (C):** The constant of proportionality between the charge Q and the potential difference U across the plates ($C = Q/U$).
- **Dielectric:** The insulating material between plates. It increases capacitance by a factor of ϵ_r (relative permittivity).
- **Breakdown Voltage (U_{max}):** The maximum potential difference that can be applied before the dielectric ionizes.

II. Mathematical Toolbox

Fundamental Equations

1. **Fundamental Law:** $Q = C \cdot U$
2. **Geometric Influence:** $C = \epsilon_r \epsilon_0 \frac{S}{d}$
 - S : Overlapping area (m^2); d : Distance between plates (m)
 - $\epsilon_0 \approx 8.85 \times 10^{-12} \text{ F} \cdot m^{-1}$
3. **Stored Energy (W):** $W = \frac{1}{2} C U^2 = \frac{1}{2} Q U = \frac{1}{2} \frac{Q^2}{C}$ (Joules)
4. **Time Constant (τ):** $\tau = R \cdot C$ (Seconds)

III. Phase Analysis: Charging & Discharging

Feature	Charging (Phase 1)	Discharging (Phase 2)
Voltage $u_C(t)$	$u_C(t) = E(1 - e^{-t/\tau})$	$u_C(t) = E \cdot e^{-t/\tau}$
Initial ($t = 0$)	$u_C = 0, i = E/R$	$u_C = E, i = -E/R$
At $t = \tau$	$u_C \approx 0.63E$	$u_C \approx 0.37E$
Steady State	$u_C \approx E$ (Open circuit)	$u_C \approx 0$ (Short circuit)

IV. Grouping & Conversions

Parallel Combination: $C_{eq} = \sum C_i$ (Increases total area S).

Series Combination: $\frac{1}{C_{eq}} = \sum \frac{1}{C_i}$ (Increases separation d).

Technical Units & Safety

Sub-multiples: $1 \mu F = 10^{-6} F$ | $1 nF = 10^{-9} F$ | $1 pF = 10^{-12} F$

Safety Note: Always discharge capacitors with a resistor before handling to avoid shock from stored energy (W).